

Conditional Statement, Switch, Loop (for, while, do-while)

- Md. Morshedul Arefin

Conditional Statement

- A conditional statement is used to make decisions in a PHP program.
- It allows the program to execute different blocks of code based on whether a condition is true or false.

if Statement

Way 1:

```
if(condition) {  
    code to be executed if condition is true;  
}
```

Way 2:

```
if(condition):  
    code to be executed if condition is true;  
endif;
```

if...else Statement

Way 1:

```
if(condition) {  
    code to be executed if condition is true;  
} else {  
    code to be executed if condition is false;  
}
```

Way 2:

```
if(condition):  
    code to be executed if condition is true;  
else:  
    code to be executed if condition is false;  
endif;
```

if...elseif...else Statement

Way 1:

```
if(condition) {  
    code to be executed if this condition is true;  
} elseif(condition) {  
    code to be executed if first condition is false and this condition is true;  
} else {  
    code to be executed if all conditions are false;  
}
```

Way 2:

```
if(condition):  
    code to be executed if this condition is true;  
elseif(condition):  
    code to be executed if first condition is false and this condition is true;  
else:  
    code to be executed if all conditions are false;  
endif;
```

Example

Example:

```
$a = 10;  
if($a == 10) {  
    echo "The value is 10"; // The value is 10  
}
```

Example:

```
$a = 10;  
if($a == 10):  
    echo "Bangladesh"; // Bangladesh  
endif;
```

Example:

```
$a = 10;  
if($a == 10) {  
    echo 'Bangladesh'; // Bangladesh  
} else {  
    echo 'USA';  
}
```

Example (Continue ...)

Example:

```
$a = 10;
```

```
if($a == 10):
```

```
    echo 'Bangladesh'; // Bangladesh
```

```
else:
```

```
    echo 'USA';
```

```
endif;
```

Example (Continue ...)

```
$age = 12;  
if($age < 18) {  
    echo "His age is {$age} and He is not adult yet";  
} elseif($age == 18) {  
    echo "His age is {$age} and he just reached in adult age";  
} else {  
    echo "His age is {$age} and he is over 18";  
}
```


Nested if, elseif, else Conditions

```
$city = "Dhaka";  
$country = "USA";  
if($country == "Bangladesh") {  
    if($city == "Dhaka") {  
        echo "This is the capital of Bangladesh.";  
    } elseif($city == "Khulna") {  
        echo "This is the city of Sundarban.";  
    } else {  
        echo "This is not a valid Bangladeshi city.";  
    }  
} elseif($country == "USA") {  
    if($city == "New York") {  
        echo "This is the city of Bangladeshi people lives in USA.";  
    } elseif($city == "Washington") {  
        echo "This is the capital of USA.";  
    } else {  
        echo "This is not a valid USA city.";  
    }  
}
```

Switch Statement

- The switch statement is used to perform different actions based on different conditions.

Syntax:

```
switch (n) {  
    case label1:  
        code to be executed if n=label1;  
        break;  
    case label2:  
        code to be executed if n=label2;  
        break;  
    case label3:  
        code to be executed if n=label3;  
        break;  
    ...  
    default:  
        code to be executed if n is different from all labels;  
}  
}
```

Example

```
$country = "USA";  
switch($country) {  
    case "Bangladesh":  
        echo "You live in Bangladesh";  
        break;  
    case "USA":  
        echo "You live in USA";  
        break;  
    case "Australia":  
        echo "You live in Australia";  
        break;  
    default:  
        echo "I do not know where you live";  
}
```

Example (Continue ...)

```
$score = 85;  
switch (true) {  
    case ($score >= 90 && $score <= 100):  
        echo "A";  
        break;  
    case ($score >= 80 && $score < 90):  
        echo "B";  
        break;  
    case ($score >= 70 && $score < 80):  
        echo "C";  
        break;  
    case ($score >= 60 && $score < 70):  
        echo "D";  
        break;  
    default:  
        echo "F";  
}
```

Example (Continue ...)

```
$country = "Greece";  
switch ($country) {  
    case "Bangladesh":  
    case "Japan":  
        echo "This is Asian Country";  
        break;  
    case "Greece":  
    case "Italy":  
        echo "This is European Country";  
        break;  
    default:  
        echo "Unknown!";  
}
```

Loop

- Loops are used to execute the same block of code again and again, as long as a certain condition is true.
- Types of Loop:
 1. for loop
 2. while loop
 3. do-while loop
 4. foreach loop

for Loop

```
for (initialization; condition; increment/decrement) {  
    code to be executed for each iteration;  
}
```

Example - Counting upto 100

```
for($i=1;$i<=100;$i+=5) {  
    echo $i.' ';  
}
```

Example - Even/Odd numbers

```
for($i=1;$i<=100;$i++) {  
    if($i%2 != 0) {  
        echo $i.' ';  
    }  
}
```

while Loop

```
while (condition is true) {  
    code to be executed;  
}
```

Example:

```
$a = 1;  
while($a < 10) {  
    echo "Bangladesh";  
    $a++;  
}
```

Example:

```
$a = 1;  
while(1) {  
    echo "Bangladesh";  
    $a++;  
    if($a > 4) { break; }  
}
```


do-while Loop

```
do {  
    code to be executed;  
} while (condition is true);
```

Example:

```
$a = 1;  
do {  
    echo "Bangladesh";  
    $a++;  
} while($a < 10);
```